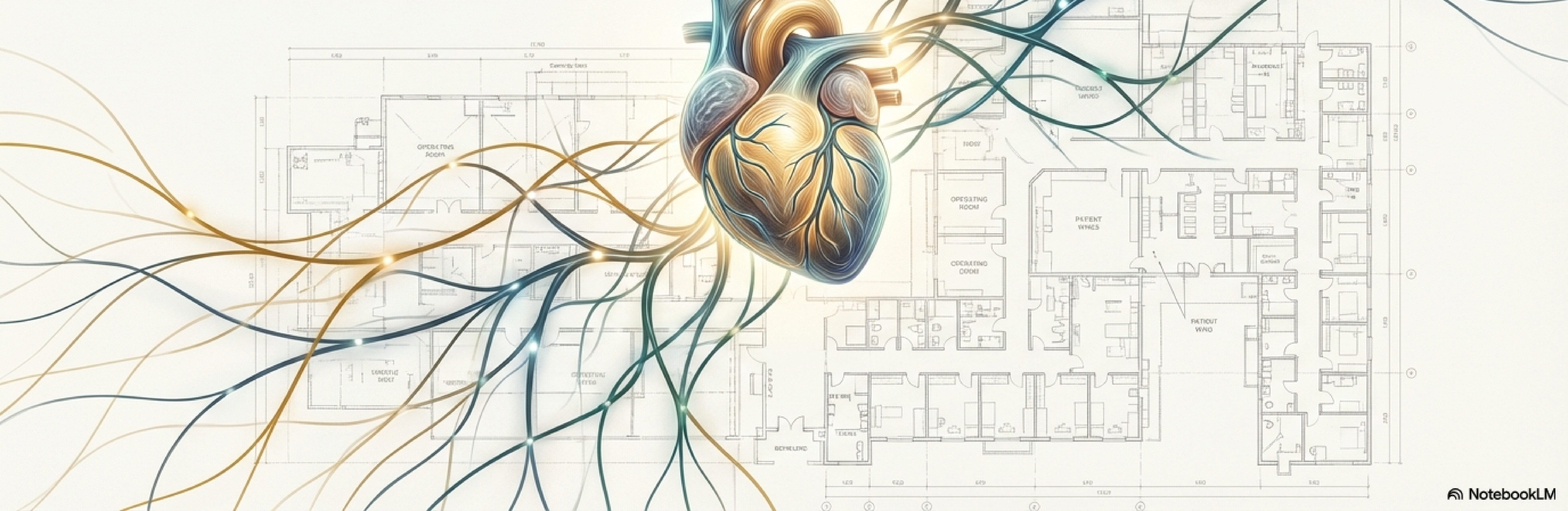
From Blueprint to Brains: Building the Digital Heart of a Smart Hospital

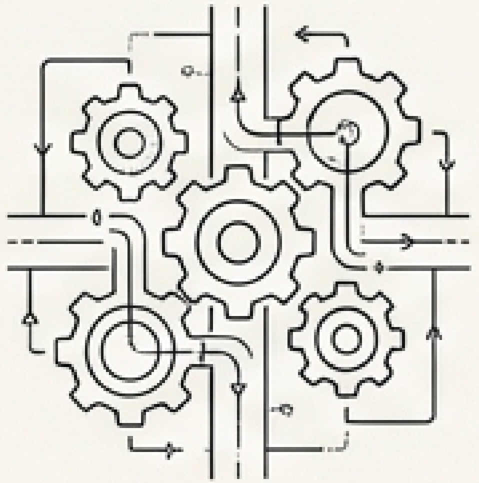
An Executive and Architectural Summary of the Multi-Agent City Information Modeling System



NotebookLM

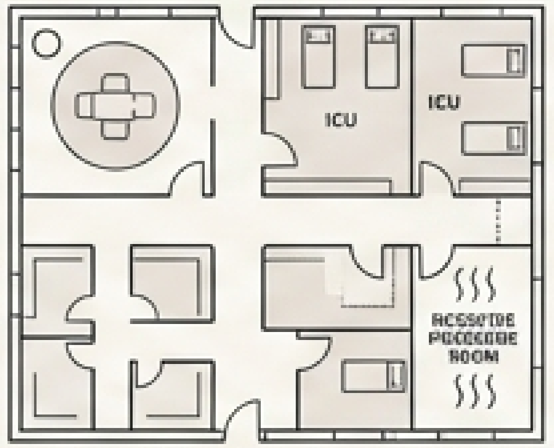
A Hospital is a City of Interconnected Systems, Not Just a Building

Managing a modern hospital requires orchestrating a complex web of life-critical systems where failures can cascade. Traditional building models are static and siloed, failing to capture this dynamic reality.



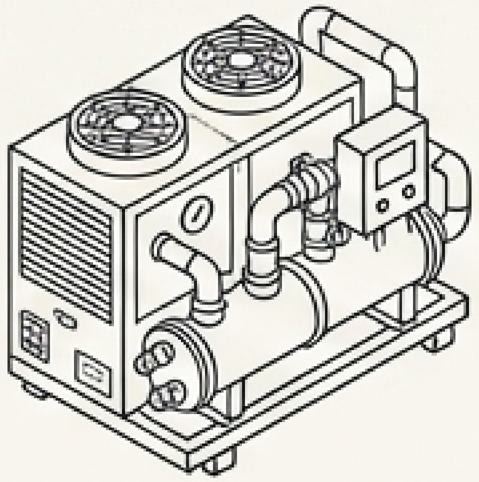
26+ Core Technical Systems

modeled across 7 major domains (HVAC, Electrical, Medical Gas, Plumbing, Fire Protection, Automation, Vertical Transport).



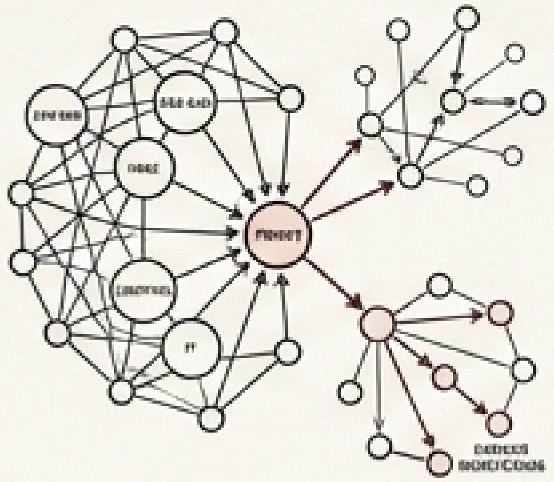
Dozens of Critical Zones

(e.g., Operating Rooms, ICUs, Negative Pressure Rooms), each with unique environmental and systemic requirements.



123+ Unique Equipment Types

from chillers to UPS systems, each with complex operational parameters and maintenance needs.

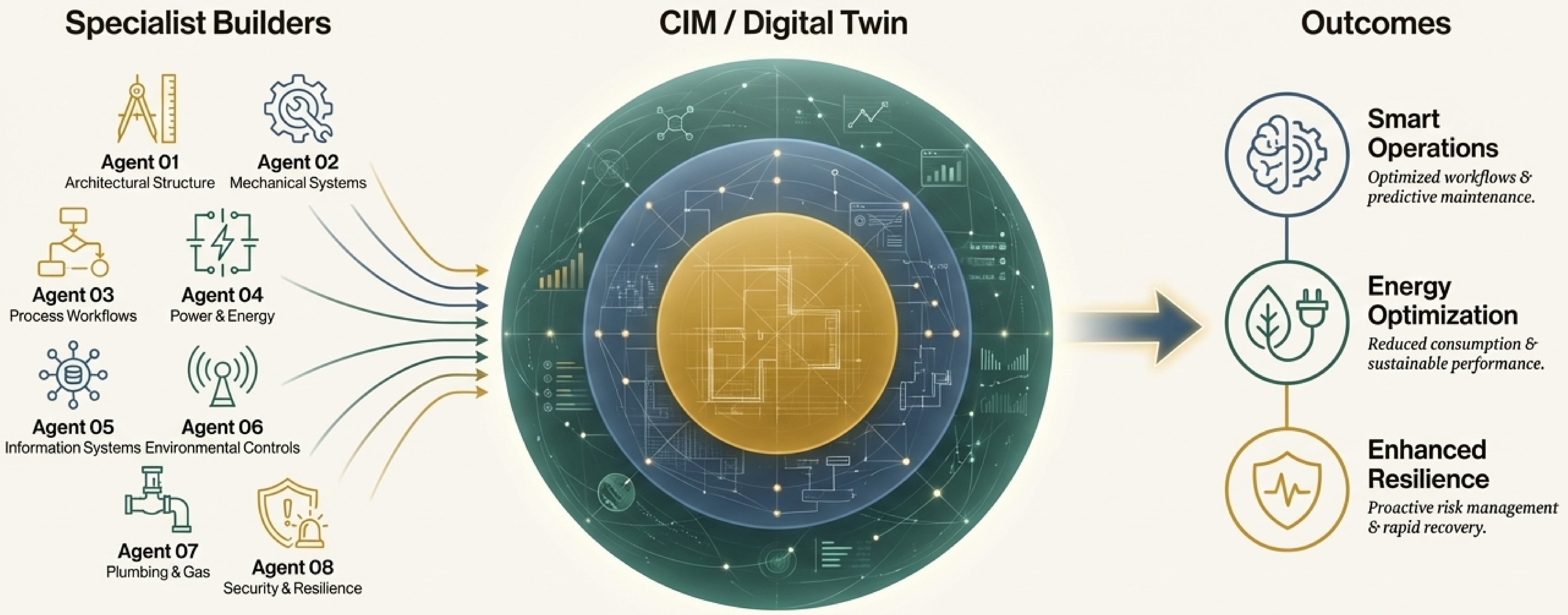


Thousands of Critical Dependencies

where a failure in one system (e.g., power) can impact dozens of others (e.g., medical gas, HVAC).

Our Solution: A Living Digital Twin, Built by a Team of Specialist Agents

We are creating a comprehensive City Information Model (CIM) – a dynamic, multi-layered digital replica that acts as the hospital's central nervous system. Instead of a monolithic approach, we deploy a team of 8 specialized AI Agents. Each Agent is an expert in its domain, systematically constructing a piece of the model, ensuring unparalleled depth, accuracy, and integration.



The Architecture is Built in Three Logical Layers

The Operational Brain (Agents 06, 07, 08)

Makes the model *actionable*. It defines *how* the hospital is controlled, measured, and managed for peak performance and safety.

Control Logic, Performance, O&M.

The Dynamic Engine (Agents 04, 05)

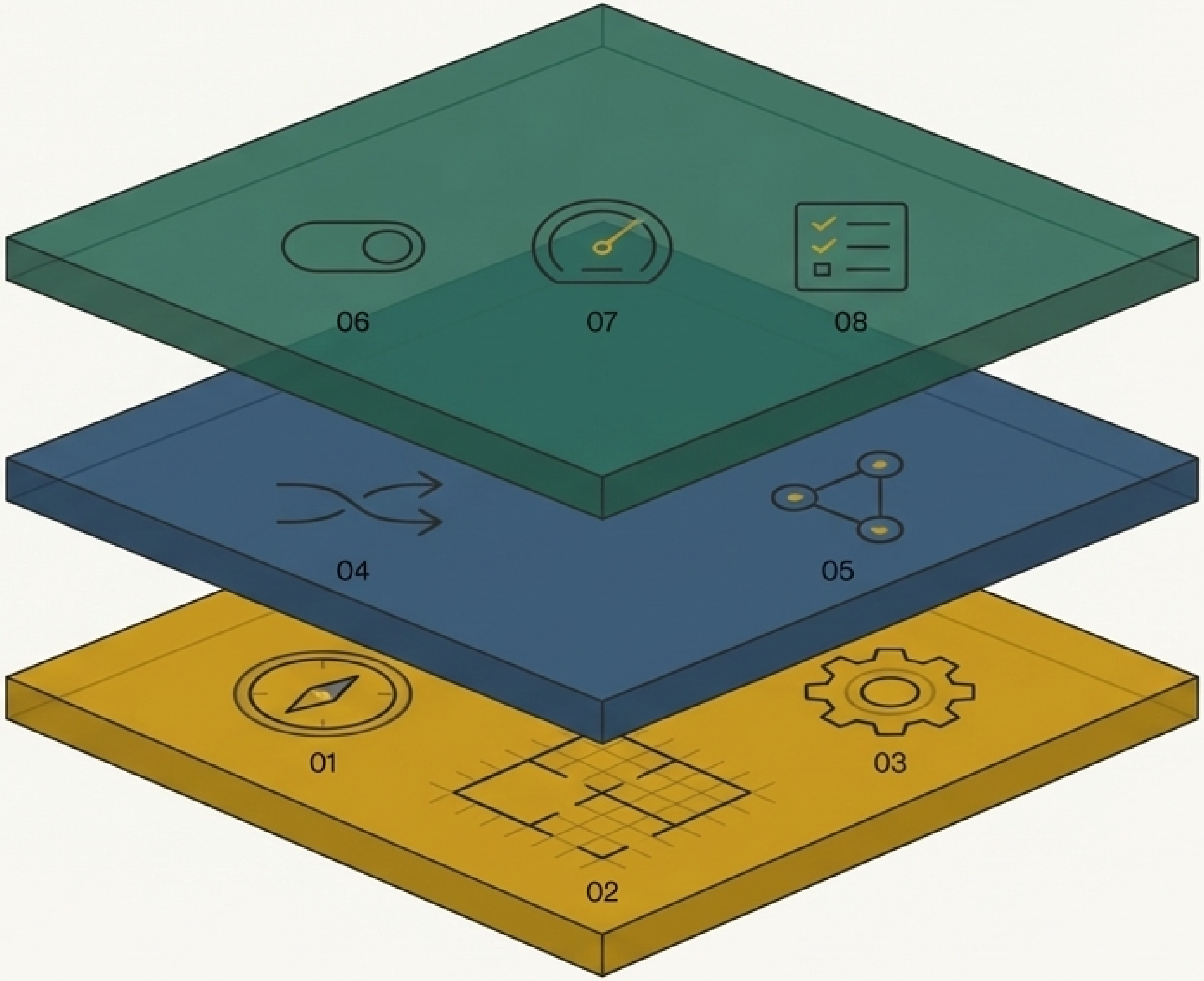
Makes the model *live*. It describes *how* things flow, interact, and influence each other, adding physics and interdependencies.

Flow Physics, System Coupling, Hypergraph.

The Digital Foundation (Agents 01, 02, 03)

Defines *what* exists and *where*. It models the static, physical reality of the hospital's systems, spaces, and equipment.

Source Serif Pro Regular
Topology, Ontology, Asset Registry.



Layer 1: The Digital Foundation Establishes a Complete Asset Model

Agents 01, 02, 03

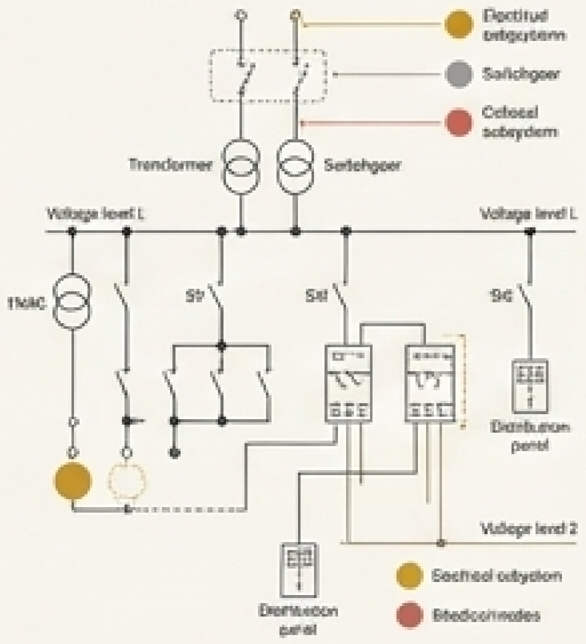
Agent 01: System Topology Architect

Role

Maps the complete "road network" of all hospital systems.

Output

26+ complete system topologies, including critical dependencies for HVAC, Electrical (HV/LV/EPS), Medical Gas (O2, VAC, AIR), and Fire Protection.



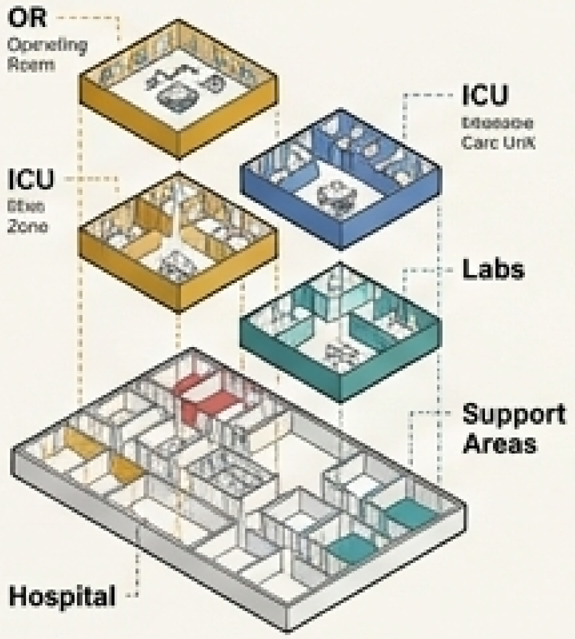
Agent 02: Space Ontology Architect

Role

Defines every space, from boiler rooms to operating rooms, codifying their purpose and requirements.

Output

A 5-level spatial hierarchy with detailed models for critical medical zones (OR, ICU, Labs), enabling "平疫结合" (Peace-to-Pandemic conversion) readiness.



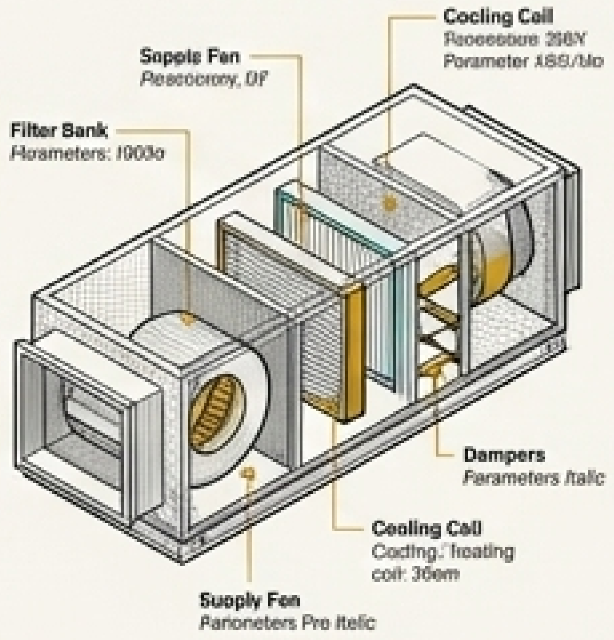
Agent 03: Equipment Ontology Architect

Role

Creates a detailed catalog of every piece of equipment, its parameters, and its function.

Output

A library of 123 equipment types, mapped to international `eClass` standards. Includes detailed models for P0-priority assets like chillers, AHUs, and UPS systems.

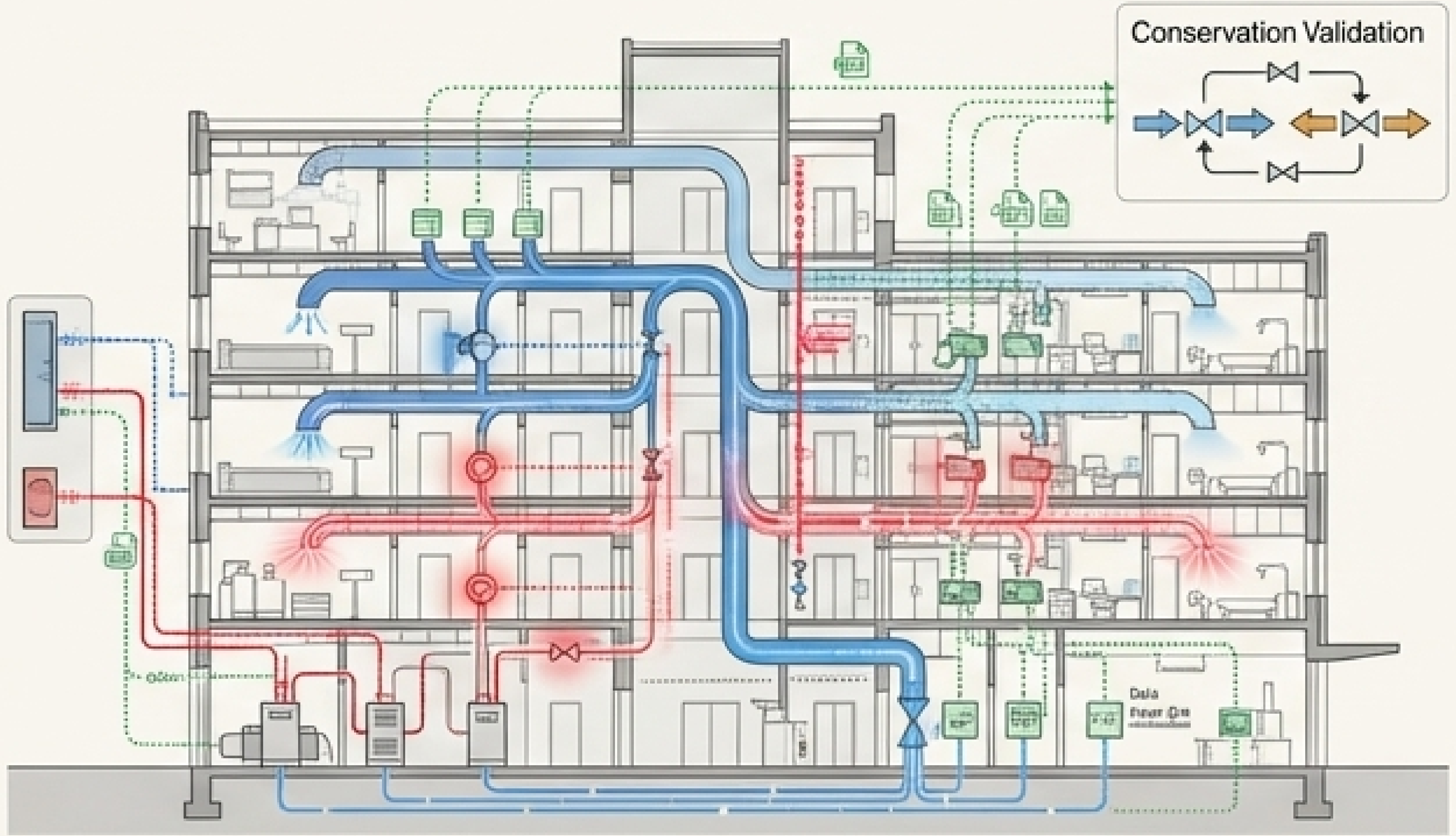


Layer 2: The Dynamic Engine Simulates the Physics of Hospital Operations

Agent 04: Flow Model Architect

Role: Models the hospital's "metabolism" – the flow of energy, matter, and information.

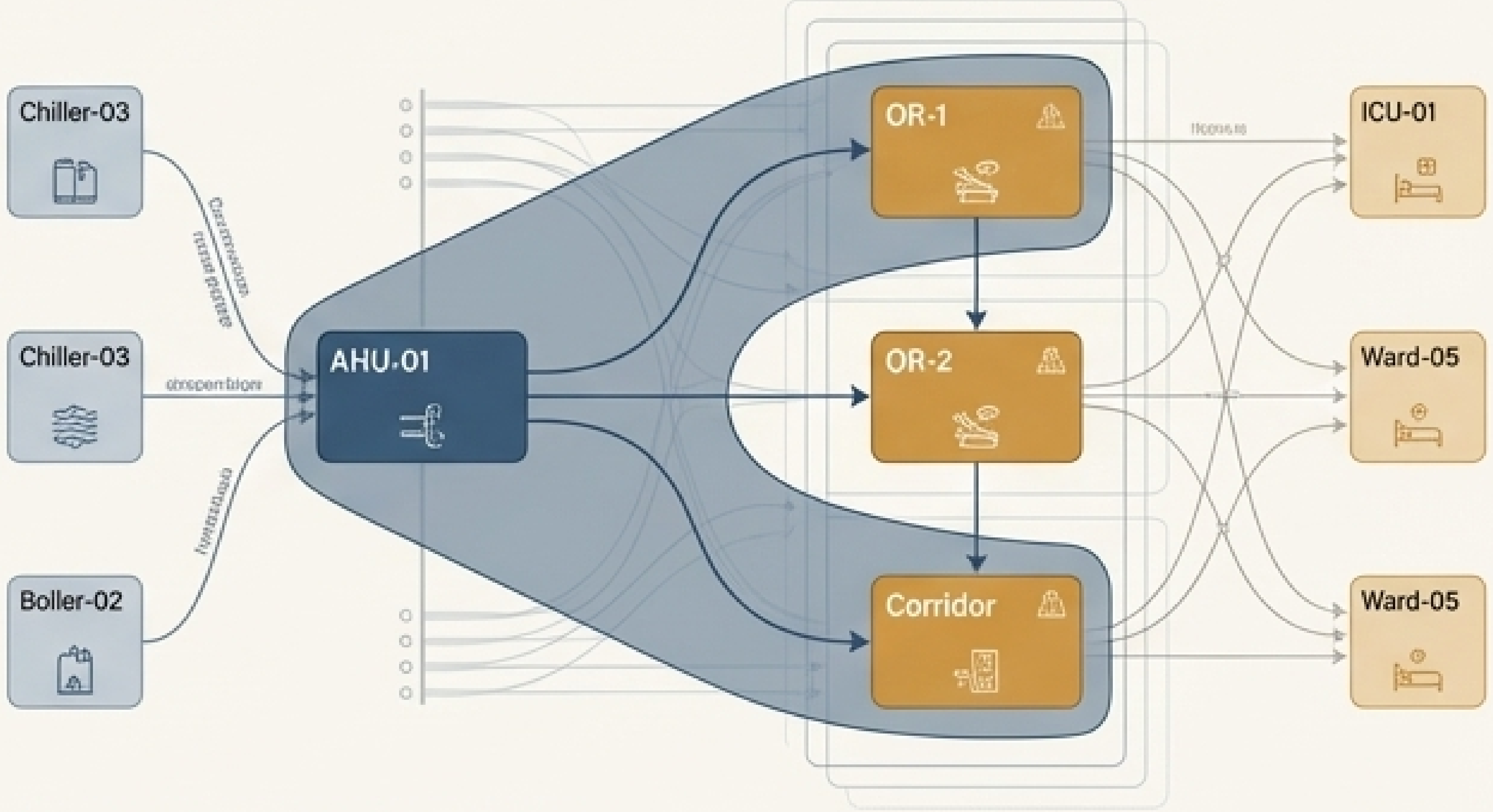
Output: A unified framework modeling 31 distinct flow types across three layers: Mass Flow (water, air), Energy Flow (electricity, heat), and Information Flow (control signals). Includes a conservation validation framework.



Agent 05: Space-System Coupling Agent

Role: The crucial integrator. Connects specific spaces to the systems that serve them, defining their complex relationships.

Output: An innovative **Hypergraph Topology** that models many-to-many relationships (e.g., one AHU serving multiple rooms). Defines **209 Coupling Units** across **28 key medical spaces** (a 40% increase in space coverage in V2.1).



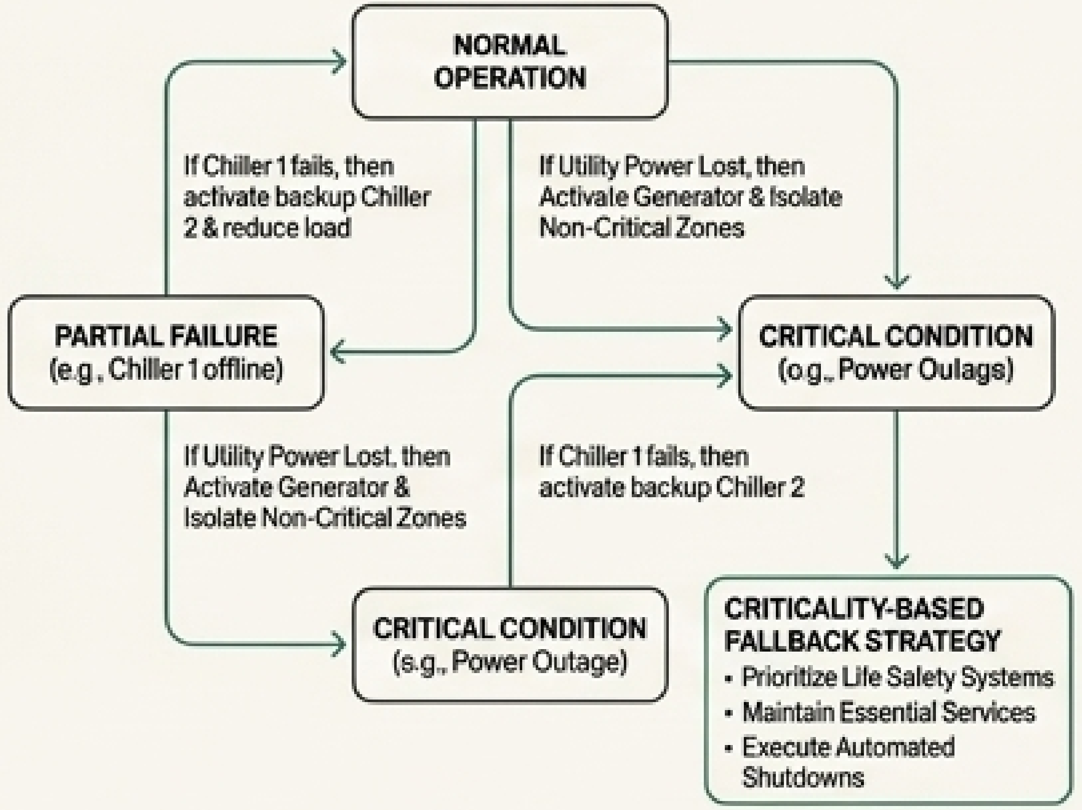
Hypergraph Topology: One-to-Many Coupling

Layer 3: The Operational Brain Drives Intelligent Management

Agent 06: Control System Architect

Role: Defines the "nervous system" – the complete control logic for the entire facility.

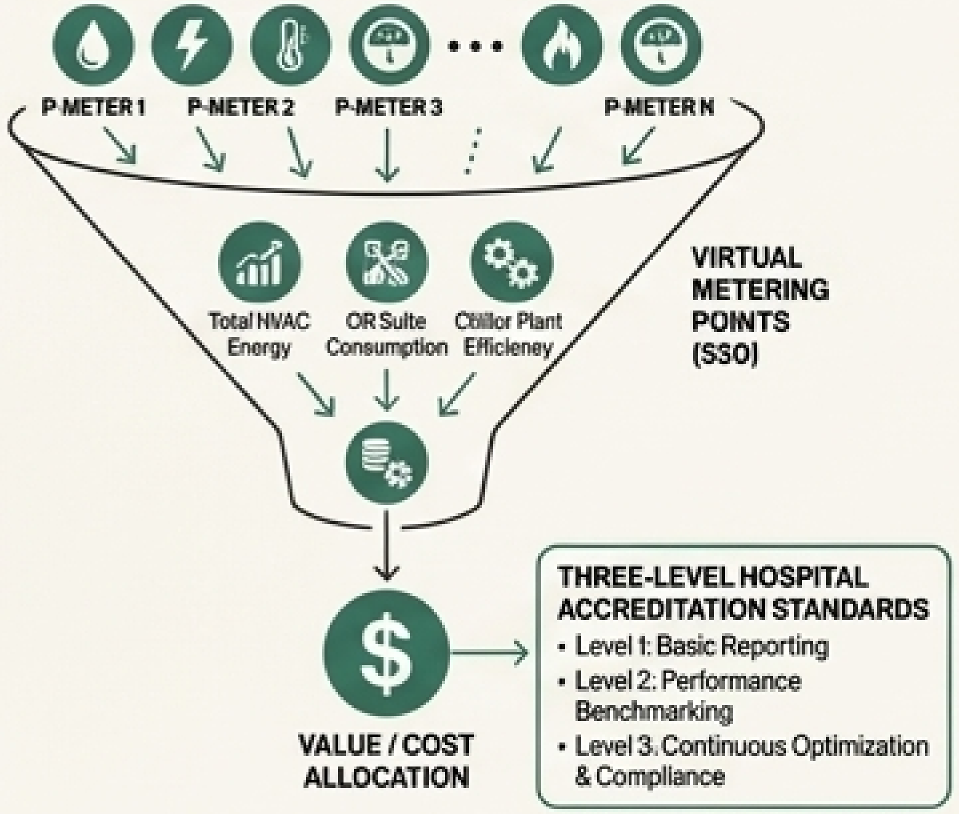
Output: Scenario-based control logic, including state machines for operating rooms and chiller plants, and a **Criticality-Based Fallback Strategy** for resilience.



Agent 07: Metering System Architect

Role: Builds the framework for performance measurement and accountability.

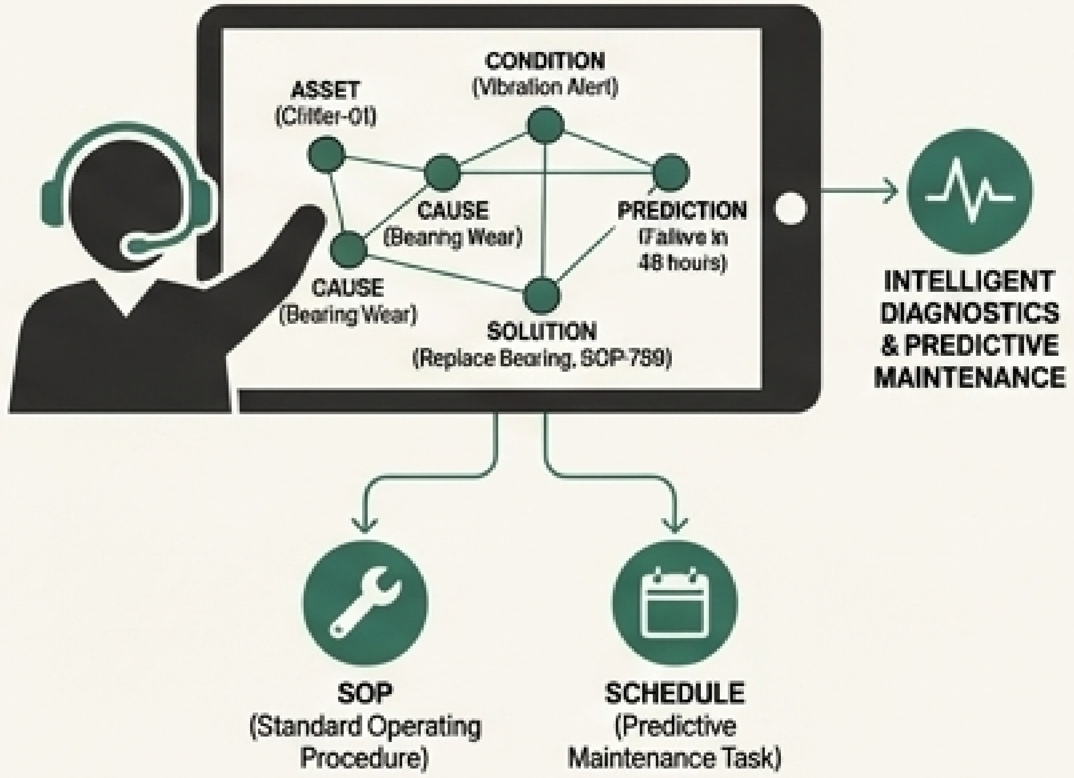
Output: A "Physical-Virtual-Value" three-layer metering network with **520 virtual metering points,** financial allocation logic, and direct mapping to the **Three-level Hospital Accreditation Standards.**



Agent 08: O&M Management Architect

Role: Models the human and procedural layer of operations.

Output: A Socio-Technical Systems (STS) model defining O&M processes, SOPs, and a knowledge graph to support **predictive maintenance** and intelligent diagnostics.



The Keystone: A Unified Framework Ensures System-Wide Cohesion

Moving from fragmented data silos to a single source of truth.

Agent 01
Site Architect

Agent 02
Facility Architect

Agent 03
Equipment Architect

Agent 04
Flow Model Architect

Agent 05
Space-System Coupling Agent

Agent 06
Control System Architect

Agent 07
Metering System Architect

Agent 08
O&M Management Architect

Integration Layer

A.1 Unified ID System

A consistent naming convention across all agents ensures every system, space, and piece of equipment has a unique, linkable identifier.

A.4 Cross-Agent Dependency Matrix

A formal matrix explicitly defines the data dependencies and relationships between all agents, ensuring model consistency.

Agent 03 v2.2 Cross-Domain Association Matrix

A powerful three-dimensional model linking System ↔ Equipment ↔ Space, enabling complex queries and impact analysis.

Agent 08 Knowledge Graph

The culmination of the integration, creating a queryable network of all entities and relationships to support advanced decision-making.

Application Deep Dive: Smart Operations and Unprecedented Resilience

From Reactive to Proactive: The CIM enables a shift from reacting to failures to predicting and preventing them.

Example 1: Scenario-Based Control (from Agent 06)

Feature: The CIM embeds detailed state machines, like the **Operating Room Environment Control State Machine**, which automatically adjusts HVAC, lighting, and pressure based on whether the room is in "Preparation," "In-Use," or "Cleaning" mode.

Benefit: Guarantees compliance, optimizes energy, and reduces manual intervention.

Example 2: Predictive Maintenance (from Agent 08)

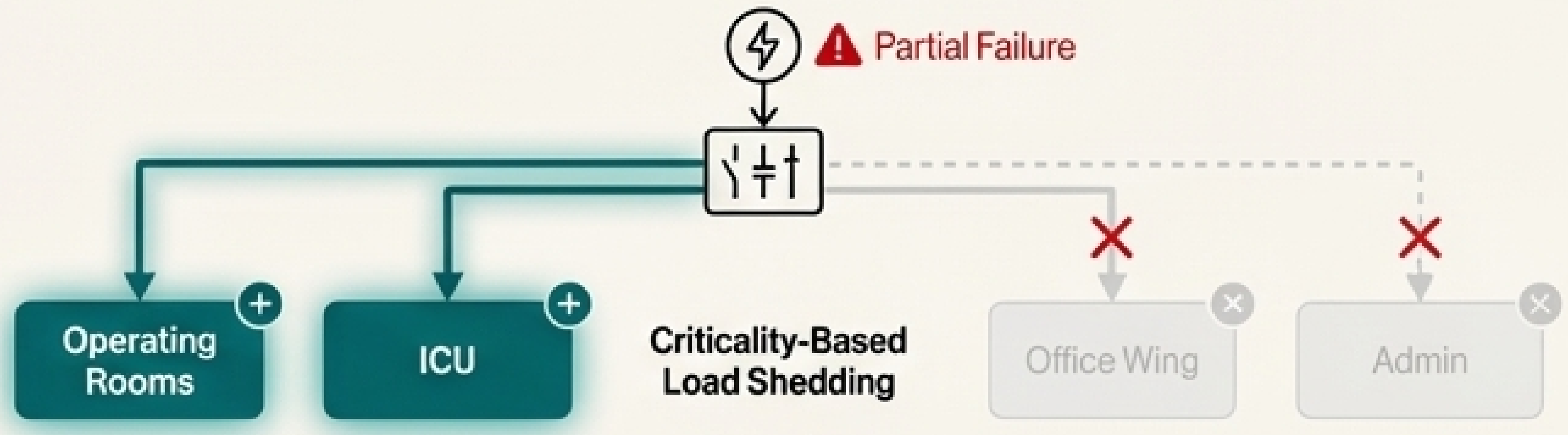
Feature: For 5 classes of critical equipment, the CIM includes specific **Remaining Useful Life (RUL) prediction models**. The system can identify subtle performance degradation (e.g., a chiller's COP dropping by 8%) and trigger a predictive maintenance work order.

Benefit: Prevents catastrophic failures, reduces downtime, and optimizes maintenance scheduling.

Example 3: Criticality-Based Resilience (from Agent 06)

Feature: The system includes a **Global Interlock Matrix** and a **Criticality-Based Fallback Strategy**. In a partial power failure, the CIM knows to shed non-critical loads while maintaining life-safety systems for ORs and ICUs.

Benefit: Ensures continuity of care during emergencies.



Application Deep Dive: Data-Driven Performance and Accreditation

From Estimation to Evidence: The CIM provides a verifiable, auditable data foundation for performance management.

The Physical-Virtual-Value Metering Framework (from Agent 07)

189 physical meters form the ground truth.

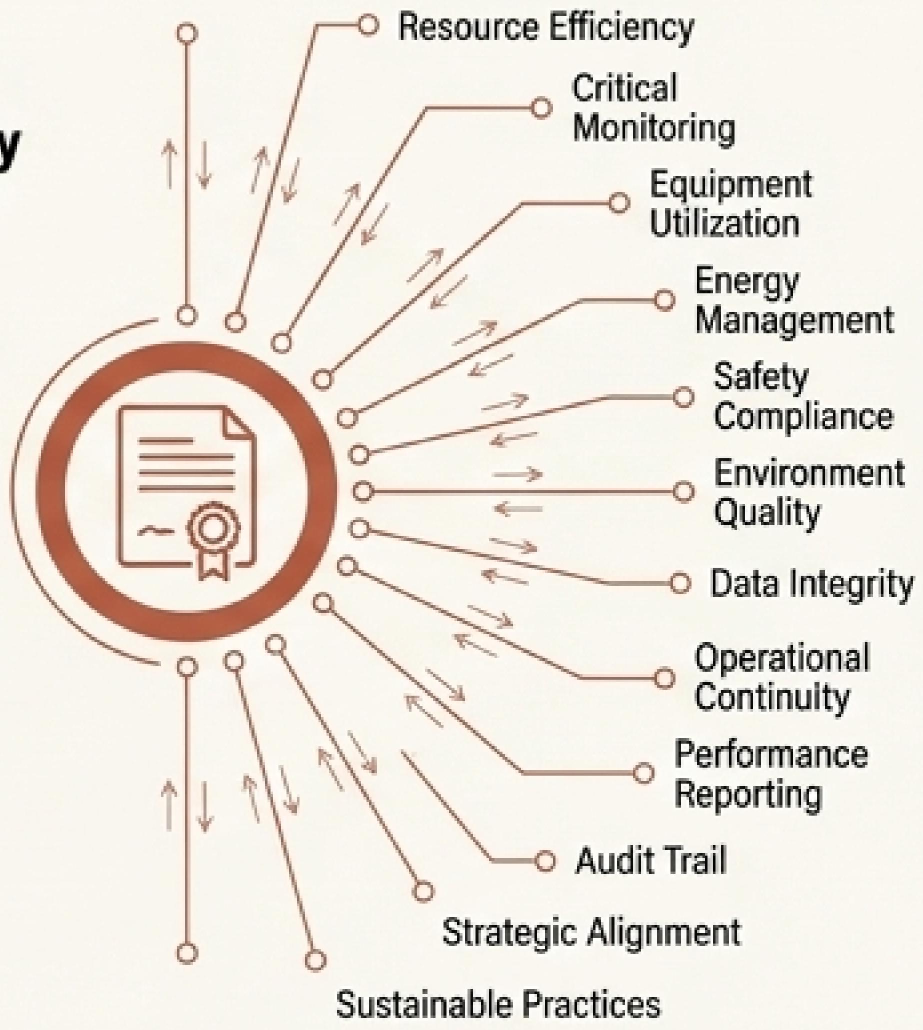
520 virtual meters, calculated using Agent-04's physics equations, provide granular cost allocation to specific departments and systems where physical meters are impractical.

24 financial allocation rules translate energy consumption into departmental costs, creating accountability.

Directly Supporting Hospital Accreditation (from Agent 07)

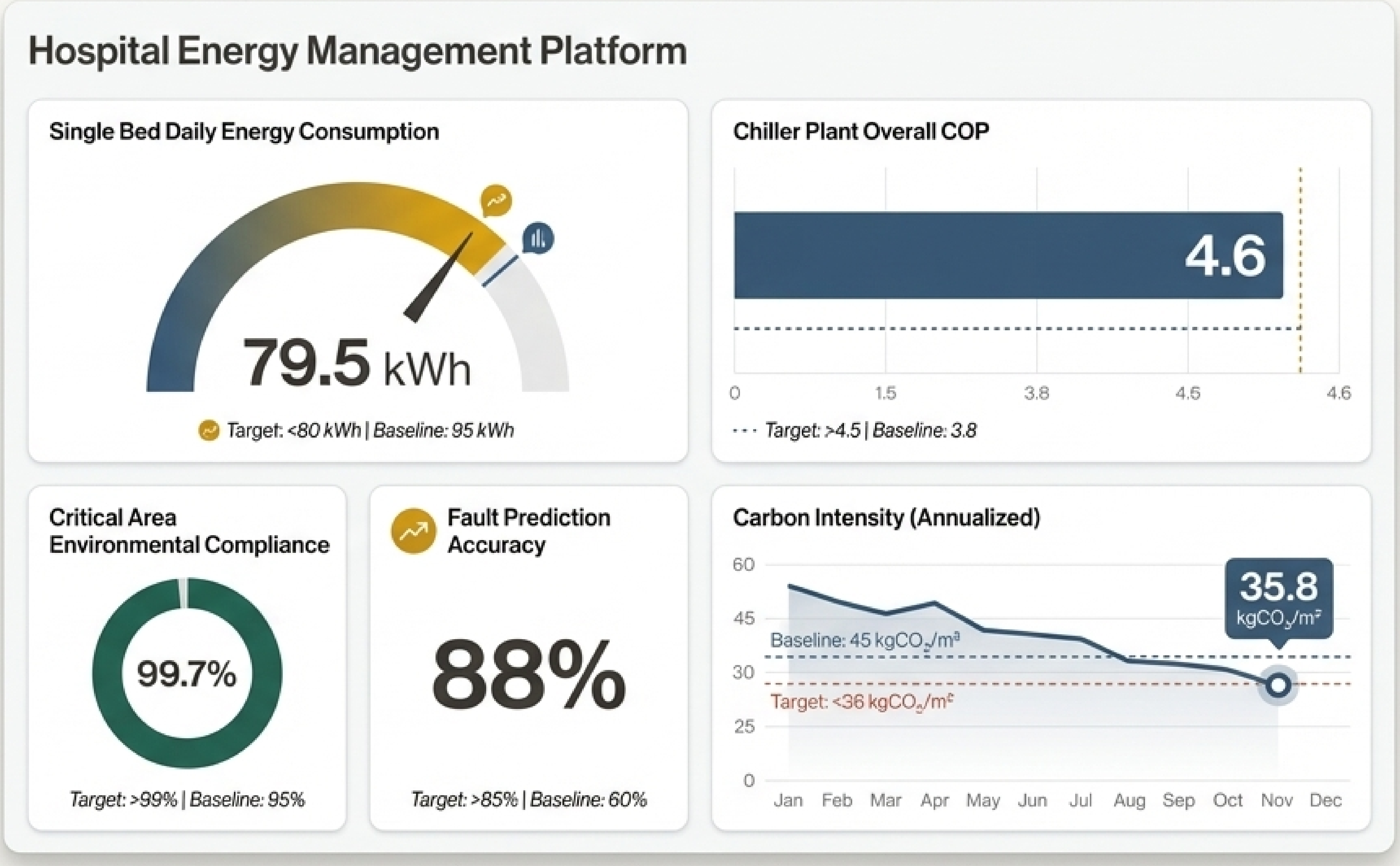
The model explicitly maps performance data to **12 key clauses** in the *Three-level Hospital Accreditation Standards (2025 Edition).*

Provides a digital evidence chain for requirements related to resource efficiency, critical department monitoring, and equipment utilization rates.



The Result: An Operations Command Center Powered by the CIM

The CIM is the data engine that powers high-level applications, transforming raw data into actionable intelligence.



A Quantifiable Leap in Model Depth, Coverage, and Quality

The multi-agent approach has produced a CIM with demonstrably superior completeness and engineering readiness.

Metric	Baseline (V1)	Current State (V2+)	Improvement	Domain
Space Coverage	20 Key Spaces	**28 Key Spaces**	+40%	Space-System Coupling (A-05)
Coupling Units	181 Units	**209 Units**	+15%	Space-System Coupling (A-05)
Equipment Standardization	70/100 Score	**95/100 Score**	**Standardization**	Equipment Ontology (A-03)
Metering Model Quality	82/100 Score	**92/100 Score**	**Implementation Ready**	Metering System (A-07)
Downstream Readiness	8.5/10 Score	**9.5/10 Score**	**Integration Ready**	Space-System Coupling (A-05)

We Haven't Just Modeled a Building; We've Built Its Central Nervous System

This multi-agent system delivers more than a static model. It creates a unified, dynamic, and intelligent digital twin that is foundational for the next generation of smart hospital operations, driving efficiency, resilience, and patient safety.